Calc 3 – Extrema & Lagrange Multipliers

Worksheet A

1. Find and classify the critical points of $f(x, y) = x^2 + y^2 - 4x - 6y + 5$.

2. Find and classify the critical points of $f(x, y) = xy(1 - x - y)$.

3. Find the absolute extrema of $f(x, y) = x^2 + y^2$ on $x^2 + y^2 \leq 4$.

4. Use Lagrange multipliers to find the max/min of $f(x, y) = x + 2y$ on $x^2 + y^2 = 5$.

5. Find the dimensions of the rectangular box of maximum volume inscribed in the ellipsoid $x^2 + 4y^2 + 9z^2 = 9$ (Lagrange).

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Worksheet B

1. Find and classify the critical points of $f(x, y) = x^3 + y^3 - 3xy$.

2. Find the local extrema of $f(x, y) = e^{x^2 - y^2}$.

3. Use Lagrange to extremize $f(x, y, z) = x + y + z$ subject to $x^2 + y^2 + z^2 = 3$.

4. Use Lagrange to find the minimum distance from the origin to the plane $x + 2y + 3z = 14$.

5. Maximize $f(x, y) = xy$ subject to $4x^2 + y^2 = 8$.

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Answer Key (Worksheets A & B)

Worksheet A.

1. $\nabla f = \langle 2x-4, 2y-6\rangle = \mathbf{0}$ at $(2,3)$. $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = 0$; $H = 4 > 0$, $f_{xx} > 0$, so **local min at** $(2,3)$, value -8.

2. $f_x = y(1-2x-y)$, $f_y = x(1-x-2y)$. Critical points: $(0,0),(1,0),(0,1),(1/3,1/3)$. At $(1/3,1/3)$: $f_{xx} = -2/3$, $f_{yy} = -2/3$, $f_{xy} = 1/3$. $H = 4/9 - 1/9 > 0$, $f_{xx} < 0$, **local max**, value $1/27$. The other three are saddle points.

3. Interior critical point only at $(0,0)$, where $f = 0$. On boundary $x^2+y^2 = 4$: $f = 4$. **Min** $= 0$ **at** $(0,0)$**; max** $= 4$ **on the circle.**

4. $\nabla f = \langle 1,2\rangle = \lambda\langle 2x,2y\rangle$, so $x = 1/(2\lambda)$, $y = 1/\lambda$. Constraint: $\frac{1}{4\lambda^2} + \frac{1}{\lambda^2} = 5$, so $\lambda^2 = 1/4$, $\lambda = \pm 1/2$. Points $(1,2)$ (**max** $f = 5$) and $(-1,-2)$ (**min** $f = -5$).

5. Maximize $V = 8xyz$ subject to $g = x^2+4y^2+9z^2 = 9$ $(x,y,z > 0)$. Lagrange equations: $8yz = 2\lambda x$, $8xz = 8\lambda y$, $8xy = 18\lambda z$. Multiplying these gives $x^2 = 3$, $4y^2 = 3$, $9z^2 = 3$, so $x = \sqrt{3}$, $y = \sqrt{3}/2$, $z = 1/\sqrt{3}$. Maximum volume $V = 8xyz = 8\cdot\sqrt{3}\cdot\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}} = \boxed{4\sqrt{3}}$.

Worksheet B.

1. $f_x = 3x^2-3y = 0 \Rightarrow y = x^2$; $f_y = 3y^2-3x = 0 \Rightarrow x = y^2 = x^4$, so $x(x^3-1) = 0 \Rightarrow x = 0$ or 1. Points: $(0,0)$ and $(1,1)$. $f_{xx} = 6x, f_{yy} = 6y, f_{xy} = -3$. At $(0,0)$: $H = 0-9 < 0$, **saddle**. At $(1,1)$: $H = 36-9 > 0$, $f_{xx} > 0$, **local min**, value -1.

2. $f_x = 2xe^{x^2-y^2}$, $f_y = -2ye^{x^2-y^2}$. Zero only at $(0,0)$. $f_{xx} = (2+4x^2)e^{x^2-y^2}$, $f_{yy} = (-2+4y^2)e^{x^2-y^2}$, $f_{xy} = -4xye^{x^2-y^2}$. At $(0,0)$: $f_{xx} = 2$, $f_{yy} = -2$, $H = -4 < 0$, **saddle point**.

3. $\nabla f = \lambda\nabla g$: $1 = 2\lambda x$, etc., so $x = y = z = \frac{1}{2\lambda}$. Constraint $3\cdot\frac{1}{4\lambda^2} = 3 \Rightarrow \lambda = \pm 1/2$. Points $(1,1,1)$ with $f = 3$ (**max**) and $(-1,-1,-1)$ with $f = -3$ (**min**).

4. Min $x^2+y^2+z^2$ s.t. $x+2y+3z = 14$. ∇: $2x = \lambda$, $2y = 2\lambda$, $2z = 3\lambda \Rightarrow x = \lambda/2, y = \lambda, z = 3\lambda/2$. Constraint: $\lambda/2+2\lambda+9\lambda/2 = 14 \Rightarrow 7\lambda = 14 \Rightarrow \lambda = 2$. Point $(1,2,3)$, distance $= \sqrt{1+4+9} = \boxed{\sqrt{14}}$.

5. $\nabla f = \lambda\nabla g$: $y = 8\lambda x$, $x = 2\lambda y$. Multiply: $xy = 16\lambda^2 xy \Rightarrow \lambda = \pm 1/4$. From $y = 2x$ (taking $\lambda = 1/4$, $x,y > 0$), constraint $4x^2+4x^2 = 8 \Rightarrow x = 1, y = 2$. Max $f = \boxed{2}$.